2021 International Conference on Information Technology (ICIT)

Machine Learning Algorithms for The Classification of Cardiovascular Disease- A Comparative Study

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2021 International Conference on Information Technology (ICIT) | 978-1-6654-2870-5/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICIT52682.2021.9491677

***Abstract*— Heart disease (cardiovascular disease) is a major human disorder that significantly affects many people's lives. Diagnosing heart disease becomes an important task to reduce its sovereignty in its early stage. Machine learning methods remain the most widely used for the classification and detection processes. This work aims to design and identify a model that best classifies cardiovascular disease and predicts the presence or absence of the disease in patients using machine learning methods with accurate predictions. Therefore, this paper compares the five most powerful machine learning platforms to classify cardiovascular disease data. The proposed five different classifiers are are support vector machine (SVM), K-nearest neighbor (K-NN), Logistic regression (LR), Decision tree (DT), and Naive Bayes (NB) for the classification of cardiovascular disease (CVD). To validate the work, the dataset was obtained from the Kaggle repository online. The algorithms' performance is analyzed, evaluated, and compared by applying various performance factors. Results indicates that support vector machine (SVM) and logistic regression (LR) methods are the most efficient for diagnosing cardiovascular disease.**

Keywords—heart disease, cardiovascular disease, machine learning, classification, comparative analysis

I. Introduction

Cardiovascular disease, also known as (heart disease) is a well-known and critical human global problem that significantly affects people. A recent study estimated that millions of deaths had occurred worldwide due to heart diseases[1]–[5], representing 31% of all world deaths. Medical evidence has revealed that certain risk factors upsurge an individual's likelihood of getting heart disease (CVD). Some of these factors, as stated by [6], [7],are unhealthy diet, use of tobacco, depression, stress, excessive use of alcohol, physical inactivity, inheritance overweight, and age. Several reports by the W.H.O have shown the rise of death due to CVD diseases mainly attributed to insufficient protective measures despite increasing risk factors.

The cumulative morbidity and mortality due to heart disease worldwide have fascinated researcher's attention to perform numerous investigations in their determination to curtail the rates. However, machine learning techniques play a very dynamic role in medical data mining for information extraction analysis. These systems have broadly used to execute medical decision systems for predictions, improved healthiness policy-making and inhibition of clinical errors, early discovery, prevention of ailments, and avoidable infirmary deaths [8]. As an intelligent system, machine learning methods can be used to understand the meaning of a data set consistently and offers a suitable output from raw data for various resolutions. These make available a way of analyzing a large dataset to find patterns and relations amongst diverse entities that are not detectable without advanced analyzing techniques [9].

Due to the high dimensional and imbalanced nature of data, standard statistical approaches have stood a significant challenge and have abridged several classification techniques unfeasible. Thus, researchers [10]–[12] have proposed several techniques to handle the inherent difficulties of high dimension in data. The difficulty of precise classification is possibly due to noisy features that are non-relevant for classification but relatively lead to the accumulation of significant errors that frequently lead to inaccurate analysis. Thus, the standard classification methods do not handle these data inadequacies, outliers, and noises leading to reduced accurate results. For this reason, there is a need to identify and apply a technique capable of solving these issues.

Nowadays, classification is a persistent problem that involves various applications. Several healthcare establishments face a significant challenge in delivering eminence services like diagnosing patients appropriately and managing treatment at reasonable expenses. Therefore, classification approaches are generally used in the medical field to classify health data into different classes according to some constraints, relatively a specific classifier [13]. In general, a classification algorithm is a function that evaluates the input features so that one class is separated into positive values and the other into negative values by the output. Therefore, classifier training is essential to identify the weights that provide the classes in the data with the most precise and best separation [14].

Motivated by the advances of numerous machine learning approaches to forecasting cardiovascular disease threat and improved classification performance, this paper contributes by performing a comparative study of machine learning algorithms and identifies the utmost effective algorithm with reasonable accuracy for classifying cardiovascular disease data. In addition to establishing the performance of different algorithms in large and minor datasets with one view, classify them appropriately, and offer information on building supervised machine learning models. We proposed five machine learning algorithms: Support vector machine (SVM), K-nearest neighbor (KNN), Logistic regression (LR), Decision tree (DT), and Naïve Bayes (NB) to efficiently evaluate their performance by using clinical data obtained from cardiovascular disease patients. The rest of the paper planned as follows: Section 2 discusses related works by other researchers. Section 3 presents the proposed methodology, dataset, and methods. At the same time, the performance measures chosen are briefly discussed in Section 4. The results are presented and discussed in section 5, while Section 6 concludes the paper and future works.

II. RELATED WORKS

In medical healthcare, classification is one of the most critical, important, and popular decision-making tools. There have been numerous computational intelligence methods that supports medical healthcare domain [15]– [17]. Some of the works done by other researchers are related to this area of research.

According to H. Ayatollahi, L. Gholamhosseini, and M. Salehi [18] in their work, they performed a comparative study between the artificial neural network (ANN) and support vector machine (SVM) approaches for classification based on the positive predictive value of cardiovascular disease. They use a medical data record from various hospitals for coronary artery disease patients. The data comprises 1324 instances, 25 attributes, and split for the algorithm to training and testing sets to a ratio of 70% and 30% correspondingly. Their experimental result shows that SVM achieves higher accuracy and better performance over the ANN.

Another work by M. F. Rabbi [19] proposed the most common classification models used in data mining. They use k-nearest neighbor (K-NN), artificial neural network (ANN), and support vector machine (SVM) using MATLAB multi-layered feed-forward back-propagation. Their work was analyzed using the heart disease Cleveland dataset, which contains 303 instances and 76 attributes from the UCI machine learning repository. After they pre-processed the dataset and performed the experiments, their experimental results disclosed that the SVM algorithm had outperformed the K-NN and ANN with a classification accuracy of 85%. In comparison, KNN achieves 82% and 73%, ANN approximately.

A. S. Ebenezer, S. J. Priya, D. Narmadha, and G. N. Sundar[20], selected ten different algorithms to classify coronary artery disease risk assessment. Artificial neural network (ANN), decision tree (DT), support vector machine (SVM), random forest (RF), CHAID, rule induction, naïve bayes (NB), k-nearest neighbor (KNN), [21] and decision stump (DS) are chosen to perform classification task, PIMA dataset was obtained from an online repository for the analysis of their work. The dataset contains 303 patients records with 14 attributes. Their results revealed that NB and SVM performed well to predict heart disease.

H. Mansoor, I. Y. Elgendy, R. Segal, A. A. Bavry, and J. Bian, [22] in their work compared the performances of classification algorithms for machine learning. They selected Random Forest (RF) and Logistic Regression (LR) techniques for predicting the risk level of heart disease in patients. They used united states National Inpatient Sample (NIS) data for 2011-2013 in their work. The LR technique delivers a better accuracy performance of 89% than the RF with 88% from their experimental analysis.

S.D. Desai, S. Giraddi, P. Narayankar, N.R. Pudakalakatti, and S. Sulegaon [23] experiment with logistic regression (LR) and back-propagation neural network (BFNN) to assess the classification technique's accuracy for heart disease prediction. They perform a comparative study of parametric and non-parametric methods in classifying heart disease. They obtained a Cleveland dataset consisting of 270 records and 13 features, out of which 11 are used and validate their analysis. A 10-fold cross-validation method was used to measure the unbiased estimate of their classification models from their experiment. Their results showed LR achieves 0.91% accuracy over BFNN with an accuracy of 0.88%.

S. I. Ayon, M. M. Islam, and M. R. Hossain [24] compared the computational intelligence techniques performances to classify heart disease. They selected seven computational intelligence techniques consisting such as decision tree (DT), k-nearest neighbor (KNN), support vector machine (SVM), deep neural network (DNN), random forest (RF), logistic regression (LR), and naïve bayes (NB). The Cleveland and Statlog heart disease datasets from the UCI repository are used to evaluate each technique's performance. Their work showed that 97% accuracy is achieved by the DNN, which is comparatively better than the other algorithms.

III. METHODOLOGY

The methodology for this study starts by collecting the dataset for classification. Classification is one of the essential tasks in data mining, the aim of which is to assign a document to one or more classes or categories. Therefore, this study determines an efficient algorithm for the classification of cardiovascular disease dataset. To achieve this, we assume that there are only two classes: the positive class with unpredictable results and the negative class without unforeseen discoveries. The methodological framework for the cardiovascular disease classification system is illustrated in Fig. 1.

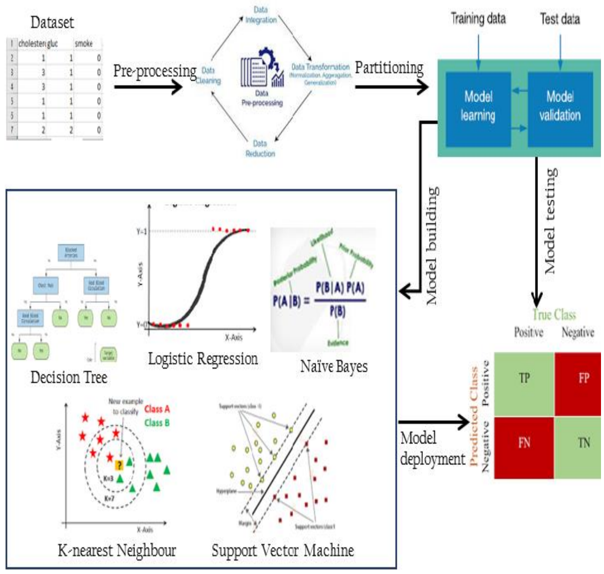

Fig. 1. Methodology framework

We use the Anaconda Jupyter Notebook Python 3 application to implement the algorithms. The datasets are then pre-processed and split into training and test sets. According to K. Korjus, M. N. Hebart, and R. Vicente, [25] most researchers practice a 70:30 ratio (70% for training and 30% for the testing) because the system generates the more data allocated to training, the more optimum and accurate outcomes. Therefore, the 70:30 partitioning ratio is performed.

A. Dataset

The dataset used is obtained from the Kaggle repository online [26] to analyze and compare the algorithms chosen for this study. The dataset consists of 77,000 clinical trial records of patients' data collected by hospitals for cardiovascular-related diseases, and there are three input features within the dataset: Objective (realistic-information), Examination (outcomes of medical investigation), and Subjective (data obtained from a patient). Also, the dataset has11 attributes from which 4 are objective features, 4 examination features, 3 subjective features, and 1 target variable labeled as (Absence or Presence) for diagnosis. Table. 1 provides a brief description of the cardiovascular disease dataset collected for the analysis of this study.

TABLE I. DATASET DESCRIPTION

	Attributes	Input Features	Data Type/ Description
1	Age	Objective features	Int / days
2	Height		Int / centimeters
3	Weight		Float/ kilograms
4	Gender		Categorical code 1: male, 2: female
5	Systolic blood pressure	Examination features	Int/
6	Diastolic blood pressure		Int/
7	Cholesterol		1: normal, 2: above normal, 3: well above normal
8	Glucose		1: normal, 2: above normal, 3: well above normal
9	Smoking	Subjective features	Binary
10	Alcohol		Binary
11	Physical activity		Binary
12	Cardiovascular	Target	Presence or absence of CVD / target variable.

B. Classification approaches

Next, we explain the machine learning approaches used in this study. Five popular classification models (i.e., Decision tree, K-nearest neighbor, Logistic regression, Naïve Bayes, and Support vector machine) are built and compared based on their predictive accuracy. Many studies compared data mining methods in different parameter settings. Most of these previous studies found these methods superior to their statistical counterparts in terms of being less constrained by assumptions and producing better classification results. Some of these methods are briefly discussed.

a) Logistic Regression (LR): is one of the most widely used machine learning models for analyzing multivariate regression issues in medical healthcare has been. LR is used to forecast a dependent variable's outcome with a continuous independent variable that helps diagnose and predict diseases differently [27]. It is an approach to discriminative categories that work on the input vector and extracts significant statistical items from the model or predicts data trends. The dependent variable in the LR is a binary variable containing data coded as 0 (yes, success, etc.) or 1. (no, failure, etc.). The main task in LR analysis is to estimate the log odds of an occurrence. LR estimates multiple linear regression functions, as defined mathematically in (1):

$$log\frac{p(y=1)}{1-(p=1)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k; \qquad (1)$$

Where k = 1, 2…, n

b) Support Vector Machine (SVM): SVM is introduced in the year 1992 by Boser, Guyon, and Vapnik initially developed as a binary classification method, and its application extended for multiple problems of classification and regression. Due to its generalization performance, it is considered a virtuous classifier without adding prior knowledge even when the input amount is very high.[28]. Data containing separable classes can be classified by finding the optimal hyperplane that maximizes the margin between classified classes[29]. SVM's model is defined as finite-dimensional vector spaces in which each dimension represents a 'feature' of a specific object. High-dimensional space problem strategy has shown to be a practical approach. In recent years, SVM has shown excellent performance for disease prediction in medical health due to its computational effectiveness on large datasets. The primary purpose is to design it as a supervised learning method for regression and classification tasks and minimize generalization errors [30]. The. SVM represented mathematically as:

$$\text{If } Y_i = +1; wx_i + b \geq 1 \qquad (2)$$

$$\text{If } Y_i = -1; wx_i + b \leq 1 \qquad (3)$$

$$\text{For all i}; y_i(w_i + b) \geq 1 \qquad (4)$$

'X' is depicted as a vector point in the Equation and "w" as a weight and a vector. Therefore, to separate the data in (2), the data in (3) must be continuously greater than zero, and the data in (4) must be continually lower than zero. Among all likely hyperplanes, SVM decides where the hyperplane distance is as great as possible.

c) K-Nearest Neighbour (K-NN): The K-NN is a method used for classification based on the similarity between one case and another. When a case is new at a particular point, its distance from every one of the model cases determined like the closest neighbor, which is the most similar the technique indicates the case. In this way, the case is put into the output containing the most immediate neighbours. The K-NN algorithm predicts a new input class label, and K-NN uses the similarity of the new input to the samples of its input in the training set. If the new input is the same as the samples in the training set, the K-NN classification output is not sound [31]. Let (x,y) be the observation of training and the learning function as h: X ⟶ Y, so that the value of y can determine by an observation x, h(x).

Mathematically, the Euclidean distance calculated using the following formula, where p:x*x=R is a function that returns the distance between the two points $x(x_x, x_i')$.

$$p(x, x^i) = |x - x^i| = \sqrt{\sum_{i=1}^{d} (x_i - x_i')^2} \quad (5)$$

d) Decision Tree (DT): is another algorithm for supervised learning; using a DT, mostly classification problems are solved. DT classifies the data based on decision rules derived from the training data by calculating the entropy and gain of information. A tree structure developed for classification purposes, and each node will represent an attribute. The root node, followed by the children's nodes will be the primary one. The leaf nodes then represent the result of the decision [32]. With continuous and categorical attributes, it performs efficiently. The population is separated into two or more comparable sets in DT based on essential predictors. For each feature, the first stage of DT is to calculate entropy. Next, the dataset split with high data gain or less entropy based on the variables/predictors. The rest of the attributes are followed by the two steps as stated.

$$\text{Entrophy}(E) = \sum_{k=1}^{l} -_{qk} \log_{2qk} \quad (6)$$

Where "l" referred to a response variable module count, "qk" is the ratio of the count of the kth class procedures to whole count of models.

$$.\text{Gain}\ (E, G) = \text{Entrophy}(E) - \sum_{\text{vlevalues}(G)} \frac{|E_{v1}|}{E} \text{Entrophy}(E_{v1}) \quad (7)$$

e) Naïve Bayes (NB): is one of the most popular data mining algorithms used for classification. It is a probabilistic model built on the Bayes theorem with a strong assumption of independence between features. NB algorithm assumes that the impact of a specific feature in a class is independent of other features. Despite its simplicity, the classifier of NB often does surprisingly well and is widely used because it often performs more sophisticated classification methods. [33].

A way of calculating a posterior probability, P(C|X), from P(C), P(X) and P(X|C), is provided by the bayes theorem. NB assumes that an independent value of other predictors is the effect of a predictor (X) value on a given class (C). This assumption is, therefore, known as conditional independence.

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \quad (8)$$

$$P(c|X) = P(x_1|c) \times P(x_2|c) \times \ldots \times P(x_n|c) \times P(c) \quad (9)$$

IV. EVALUATION MEASURES

The evaluation criteria used to measure the algorithm's performance are based on specific metrics such as f1_score, precision, recall, and accuracy. We also measure the training time taken for each algorithm. The confusion matrix presents the classification algorithm's performances that form the basis from which various parameters are calculated. Thus, the confusion metrics measure a model's accuracy by comparing predicted values with actual values and contribute by finding out whether a classification algorithm is usually mislabeling one another [34], [35]. The parameters below are a brief description of values for the confusion matrix and its representation, as shown in Table 2.

- True positive (TP) – Are situations when the actual class of datapoint is 1 and predicted also 1.
- False positive (FP) – Are situations when the actual class of datapoint is 0 and predicted is 0.
- False negative (FN) – Are situations when actual class of datapoint is 0 and predicted is 1.
- True negative (TN) – Are situations when the actual class of datapoint is 1 and predicted is 0.

TABLE II. CONFUSION MATRIX

		Actual value	
		Classified as absence	***Classified as presence***
Predicted value	*Absence*	TP	FN
	Presence	FP	TN

A. F1_score

F1-score is a function interpreted as a weight of recall average and precision when an f1-score reaches its best value at 1 and its worst score at 0. For f1-score, the formula is:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

B. Recall

Recall measures how much relevant data from any machine learning algorithm is retrieved. It focuses on the capability to find all related occurrences in the data. The following equation represents a recall:

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (11)$$

C. Precision

Precision is the fact of being accurate and correct. Precision gives the idea of correctly predicted instances. It quantifies predictions that belongs to a positive class, measured the amount of true positive from all positives, and this is calculated as:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (12)$$

D. Accuracy

Accuracy Is an essential measure of describing the performance of an algorithm. It defines the step to which an algorithm can predict the positive and negative cases correctly and is measured using the formula:

$$\text{Accuracy} = \frac{TP+TN}{TP + FP + FN+TN} \quad (13)$$

V. RESULTS AND DISCUSSION

The experiments for this study are performed on intel ® core2 Duo T8300 @ 2.40 GHz Windows 10 computer with

8GB RAM and implemented in Anaconda Jupyter notebook python version 3. We validate our experiments on a single dataset from the Kaggle repository for proper understanding.

In this paper, five computational intelligence methods: decision tree (DT), logistic regression (LR), K-nearest neighbor (KNN), Naïve bayes (NB), and Support vector machine (SVM), has used for the classification of cardiovascular disease dataset. The dataset consists of 70,000 samples and 11 features. There are two categories of analyzing the classes: absence or presence of the disease. From the data samples, 35021 are specified as absent, while the remaining 334979 specified as presence of cardiovascular disease. The data is partitioned into training and testing sets at a ratio of 70:30, respectively.

The confusion matrix of prediction results for DT, LR, KNN, NB, and SVM is shown in tables 1-5. With the help of f1_score, recall, precision, and accuracy are measured as illustrated in Fig. 2-6.

TABLE III. CONFUSION MATRIX FOR DECISION TREE CLASSIFIER

		Predicted Value		
		Absence	Presence	Actual value
Actual value	Absence	6562 (31.35%)	3899 (18.46%)	10461
	Presence	3775 (18.03%)	6764 (32.16%)	10539
	Total Predicted	10337	10663	21000

TABLE IV. CONFUSION MATRIX FOR K-NEAREST NEIGHBOR CLASSIFIER

		Predicted Value		
		Absence	Presence	Actual value
Actual value	Absence	8243 (39.25%)	2296 (10.93%)	10539
	Presence	4031 (19.20%)	6430 (30.62%)	10461
	Total Predicted	12274	8726	21000

TABLE V. CONFUSION MATRIX FOR LOGISTIC REGRESSION CLASSIFIER

		Predicted Value		
		Absence	Presence	Actual value
Actual value	Absence	5363 (38.31%)	1625 (11.61%)	6988
	Presence	2244 (16.03%)	4768 (34.06%)	7012
	Total Predicted	7607	6393	14000

TABLE VI. CONFUSION MATRIX FOR NAÏVE BAYES CLASSIFIER

		Predicted Value		
		Absence	Presence	Actual value
Actual value	Absence	9078 (43.23%)	1383 (6.59%)	10461
	Presence	7134 (33.97%)	3405 (16.21%)	10539
	Total Predicted	16212	4788	21000

TABLE VII. CONFUSION MATRIX FOR SUPPORT VECTOR MACHINE CLASSIFIER

		Predicted Value		
		Absence	Presence	Actual value
Actual value	Absence	5671 (40.51%)	1317 (9.41%)	6988
	Presence	2509 (17.92%)	4503 (32.16%)	7012
	Total Predicted	8180	5820	14000

From the confusion matrix results, the naïve bayes (NB) predicts the highest number of true positives, while logistic regression predicts the highest number of true negatives, as represented in Tables 6 and 5.

As illustrated in Figure 2, is a comparison between the algorithms DT, KNN, LR, NB, and SVM for the F1_score with 63.94%, 67.02%, 71.13%, 44.43%, and 70.71%, respectively. It is concluded that LR outperforms other algorithms for the F1_score.

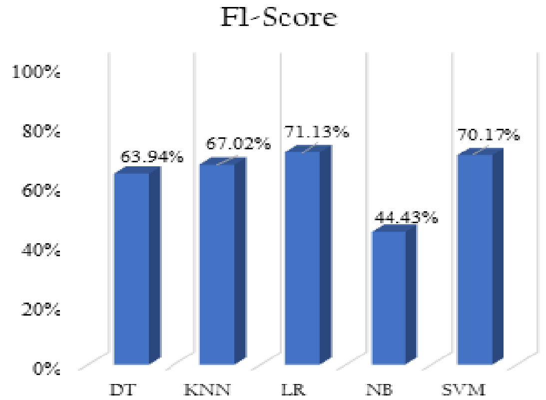

Fig. 2. Comparison for F1_score

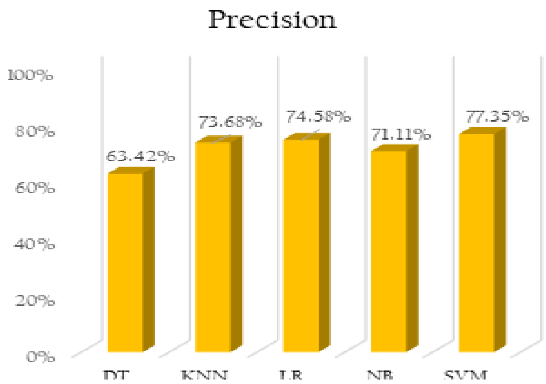

Fig. 3. Comparison for the precision

The representation in Fig. 3 shows the comparison for precision which computes the number of positive class predictions that essentially fit the positive class. The SVM outperforms other algorithms with 77.35%, followed by LR 74.58%, KNN 73.68%, NB 71.11%, and DT with 63.42%.

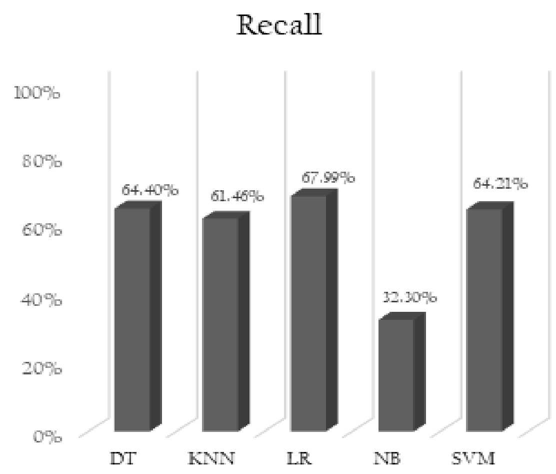

Fig. 4. Comparison of Recall

Fig. 4 shows a comparison for a recall measure between the algorithms, which illustrates the proportion of predicted positive classes made out of entire examples that are positive from the dataset.

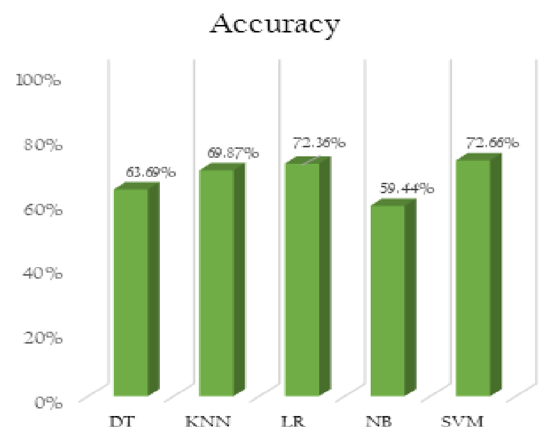

Fig. 5. Comparison for the accuracy performance

The comparison of performance accuracy for the algorithms used is shown in Fig. 5. It is observed that SVM achieved an accuracy of 72.66% over other algorithms DT, KNN, LR, and NB with 63.69%, 69.87%, 72.36%, and 59.44%.

Fig. 6. Comparison of training time

Fig. 6 represent the time taken during the training process for each of the algorithm. SVM takes a longer training time of about 296.67 seconds, despite the excellent accuracy performance over other algorithms.

In general, SVM and logistic regression algorithms have the highest classification accuracy of 72.66% and 72.33%. In contrast, DT, KNN, and NB have a classification accuracy of 63.69%, 69.87%, and 59.44%, respectively, as illustrated in Table 8. Comparing the overall results of the classification shows a support vector machine (SVM), and logistic regression have been identified as the most efficient classification algorithms for cardiovascular disease data with an accuracy of 72.66% and 72.36.

TABLE VIII. COMPARATIVE ANALYSIS FOR THE ALGORITHMS

Algorithm	Recall	F1_score	Precision	Accuracy	T/time (Sec)
DT	64.40	63.94	63.42	63.69%	0.53
K-NN	61.46	67.02	73.68	69.87.%	5.78
LR	67.99	71.13	74.58	72.36%	2.52
NB	32.30	44.43	71.11	59.44%	0.63
SVM	64.21	70.17	77.35	72.66%	296.67

Therefore in this paper, we have discussed the major techniques used in data mining to classify cardiovascular disease data. These contribute by identifying the support vector machine (SVM) as the best performing technique that can predict the presence or absence of cardiovascular disease with better accuracy for the early diagnosis, which will reduce the rate of mortality. Thus, the system helps not only doctors but also the patients by reducing the cost of laboratory examination and saves time.

VI. CONCLUSION AND FUTURE WORK

Data mining applications are widely used in medical healthcare to detect diseases and diagnose heart disease patients based on their medical data. In this paper, we have discussed effective machine learning techniques and identified the most efficient for the classification of cardiovascular disease by using the patient's data. Multiple classification algorithms SVM, KNN, DT, LR, and NB have been compared based on evaluative measures such as precision, recall, f1-score, accuracy, and the algorithms training time. Our proposed work shows that support vector machine (SVM) and logistic regression (LR) methods are the most efficient for diagnosing cardiovascular disease. In the future, we intend to enhance the performance of these basic classification techniques by developing a meta-model that will be used for predicting cardiovascular disease among people with risk of heart disease.

VII. ACKNOWLEDGMENT

The authors are thankful to the School of Computer Sciences, and Division of Research & Innovation, USM for providing financial support from the Short-Term Grant (304/PKOMP/6315435).

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